Jin-Soo Kim
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Systems Software &
Architecture Lab.

Seoul National University

Fall 2026

4190.307:
Operating Systems



Course Information

- Schedule
  - 11:00 – 12:15 (Tuesday & Thursday)
  - Lecture room: Engineering Bldg. #301-203
  - 3 credits
  - Official language: Korean

- TA: Injae Kang and Jiho Jeon (snucsl.ta [at] gmail.com)
- SNU eTL system for exam/project scores
- https://csl.snu.ac.kr/courses/4190.307/2026-2 for announcements and lecture slides
- https://sys.snu.ac.kr for project submissions and grading

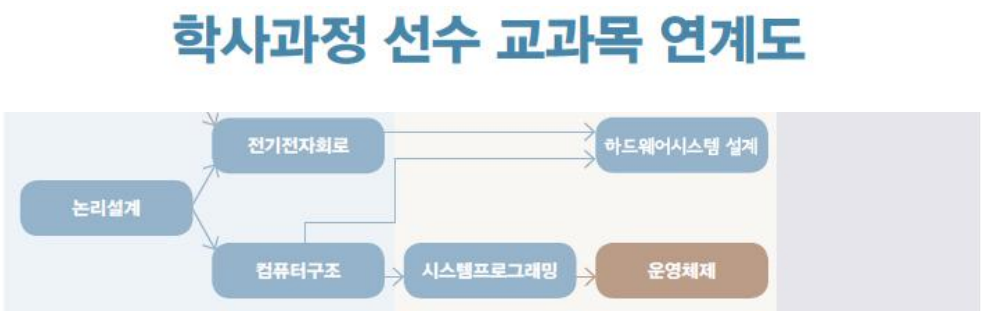
About Me



- Jin-Soo Kim (김진수)
  - Professor @ CSE Dept.
  - Systems Software & Architecture Laboratory
  - Operating systems, storage systems, parallel and distributed computing, embedded systems, …
- E-mail: jinsoo.kim@snu.ac.kr
- Tel: 02-880-7302
- Office: Engineering Bldg. #301-504
- Office hours: Tuesday & Thursday (appointments by email)
- https://csl.snu.ac.kr/

Prerequisites

- Courses
  - Computer Architecture (4190.308) – Must!
  - System Programming (M1522.000800) – Must!

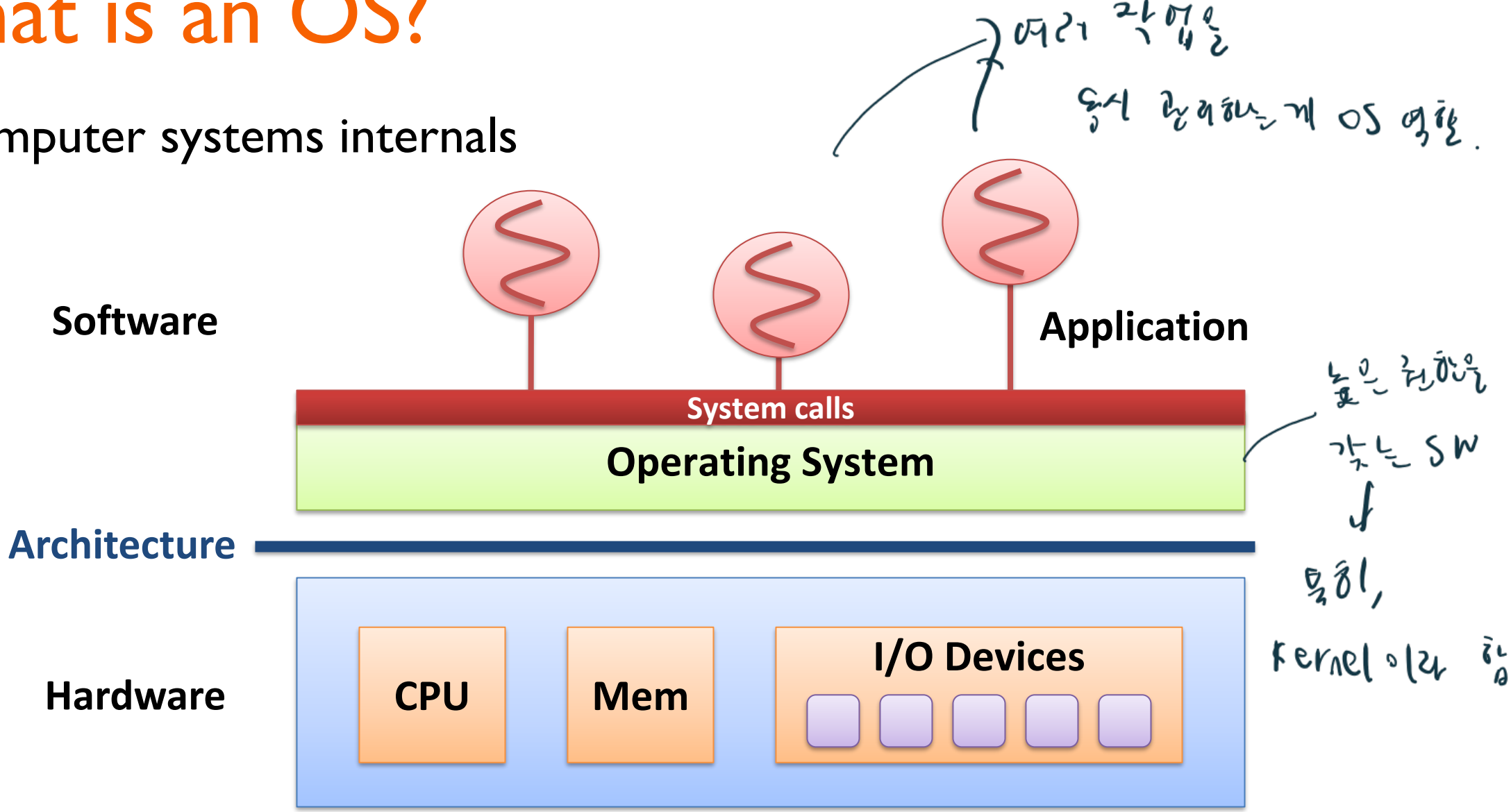


- Skills
  - Fluent C programming
  - Familiarity with Linux commands and build environment (e.g., gcc, gdb, make, …)
  - Reading a large, complex program
  - RISC-V architecture & assembly programming

- Accessible Linux (Ubuntu 24.04.1 LTS or later) or MacOS machine

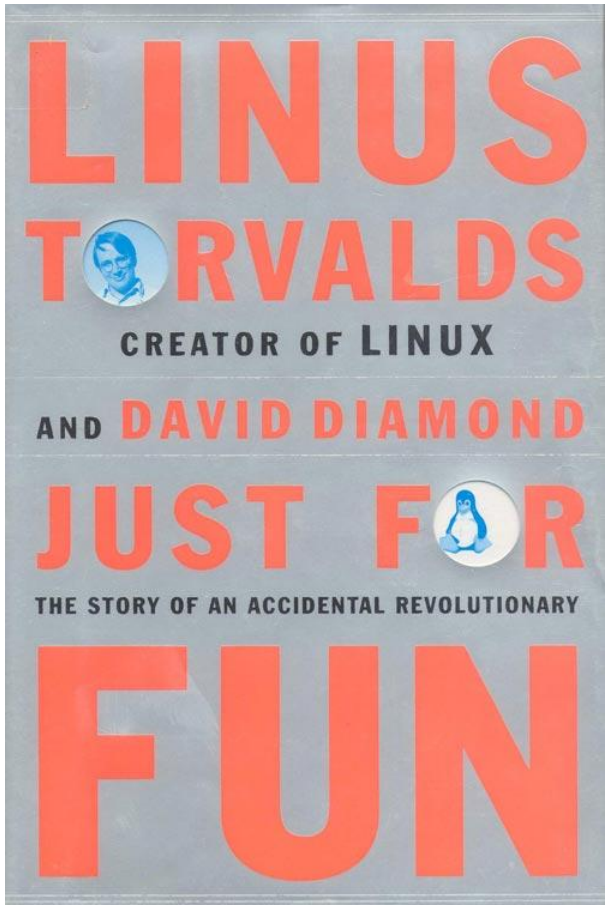
What is an OS?

- Computer systems internals



Why do we learn OS?

- ~~To graduate~~
- To understand computer systems better
- To make a better OS or system
  - Functionality
  - Performance/cost
  - Reliability
  - Energy efficiency
- To make new hardware up and running
- To design OS-aware hardware – 550
- Just for fun!

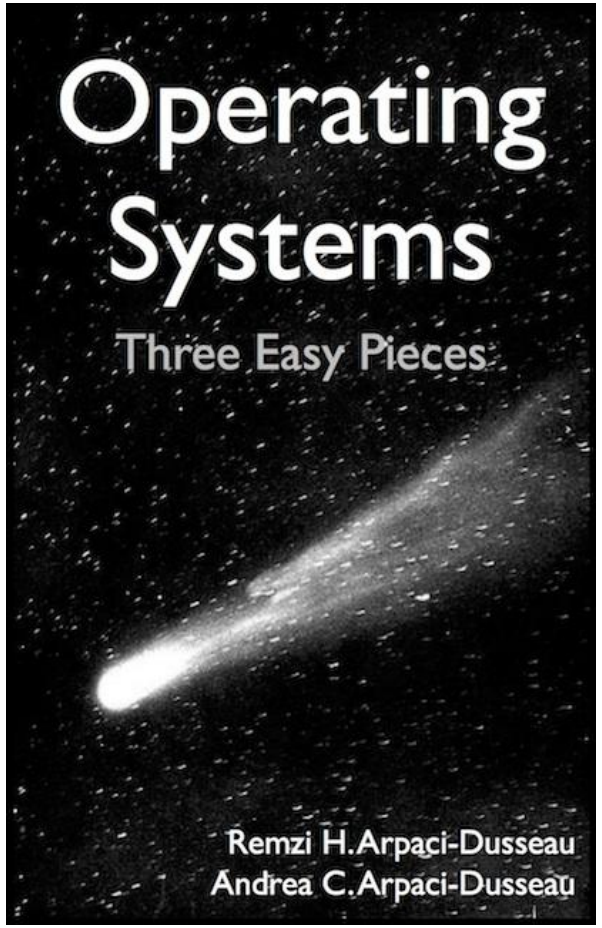


Textbook

- Operating Systems: Three Easy Pieces
  - Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau
  - Arpaci-Dusseau Books
  - November 2023 (Version 1.10)
  - Available (with several options) at http://ostep.org

  - Korean version (based on Version 0.91) is also available at https://github.com/remzi-arpacidusseau/ostep-translations/, but I highly recommend you read the original English version

  - Read Remzi's great article at http://from-a-to-remzi.blogspot.com/2014/01/the-case-for-free-online-books-fobs.html

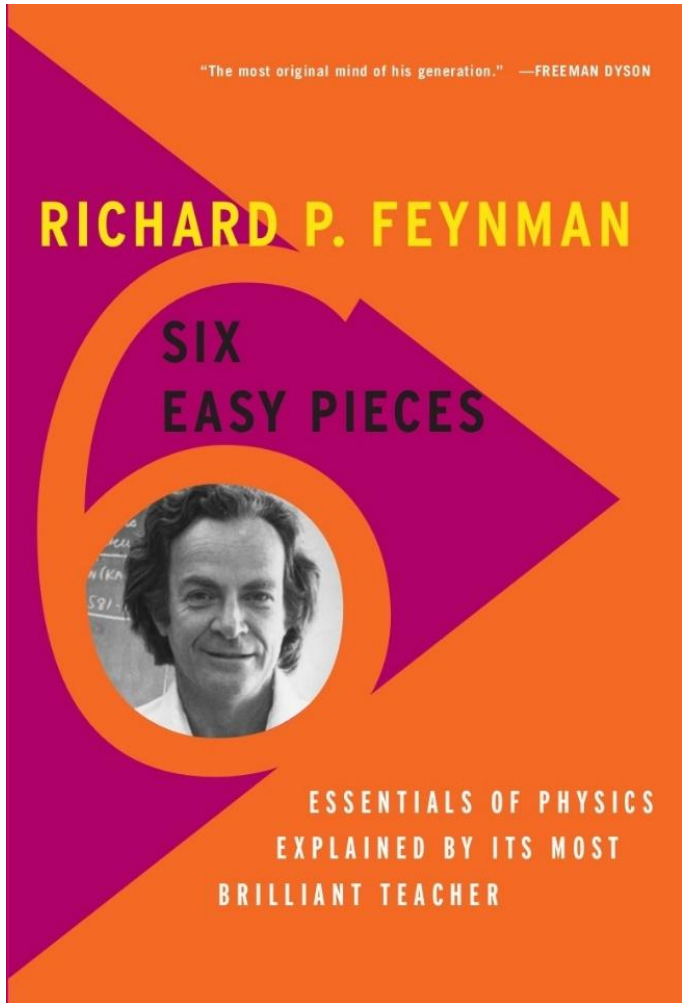


Why Three Pieces?

"… as *Operating Systems are about half as hard as Physics.*"

Chap. 1

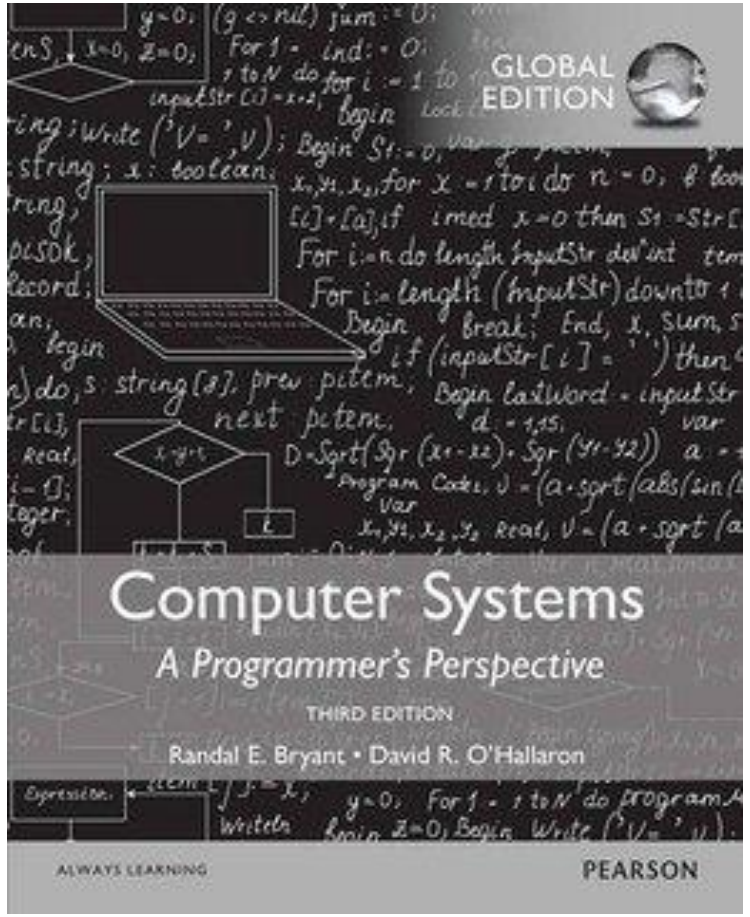
A Dialogue on the Book



Reference: CSAPP

- Computer Systems: A Programmer's Perspective
  - Randel E. Bryant and David R. O'Hallaron
  - Third Edition
  - Pearson
  - March 2015

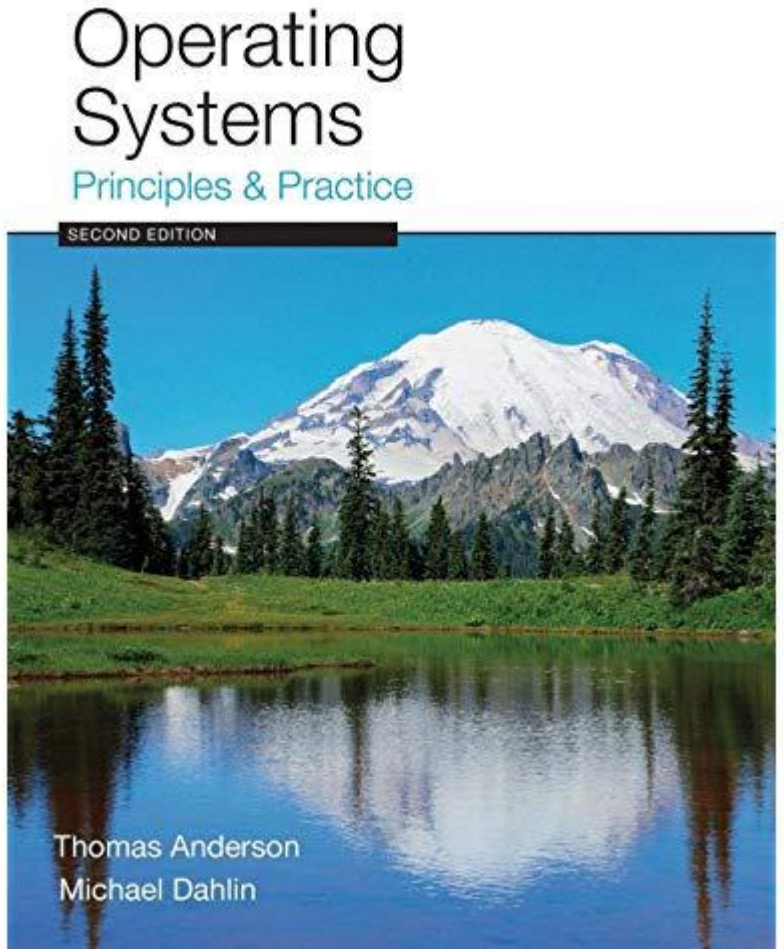
  - http://csapp.cs.cmu.edu



Reference: OSPP

- Operating Systems: Principles and Practice
  - Thomas Anderson and Michael Dahlin
  - Second Edition
  - Recursive Books
  - August 2014

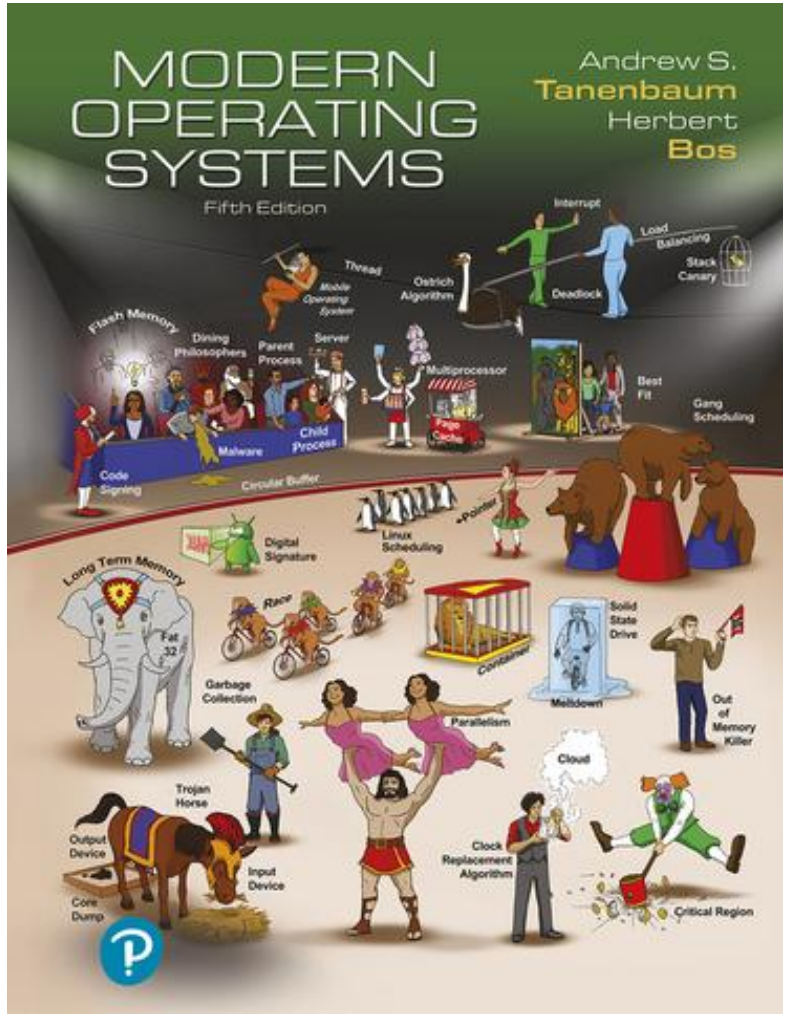
- http://ospp.cs.washington.edu/



Reference: MOS

- Modern Operating Systems
  - Andrew S. Tanenbaum and Herbert Bos
  - Fifth Edition
  - Pearson
  - October 2022

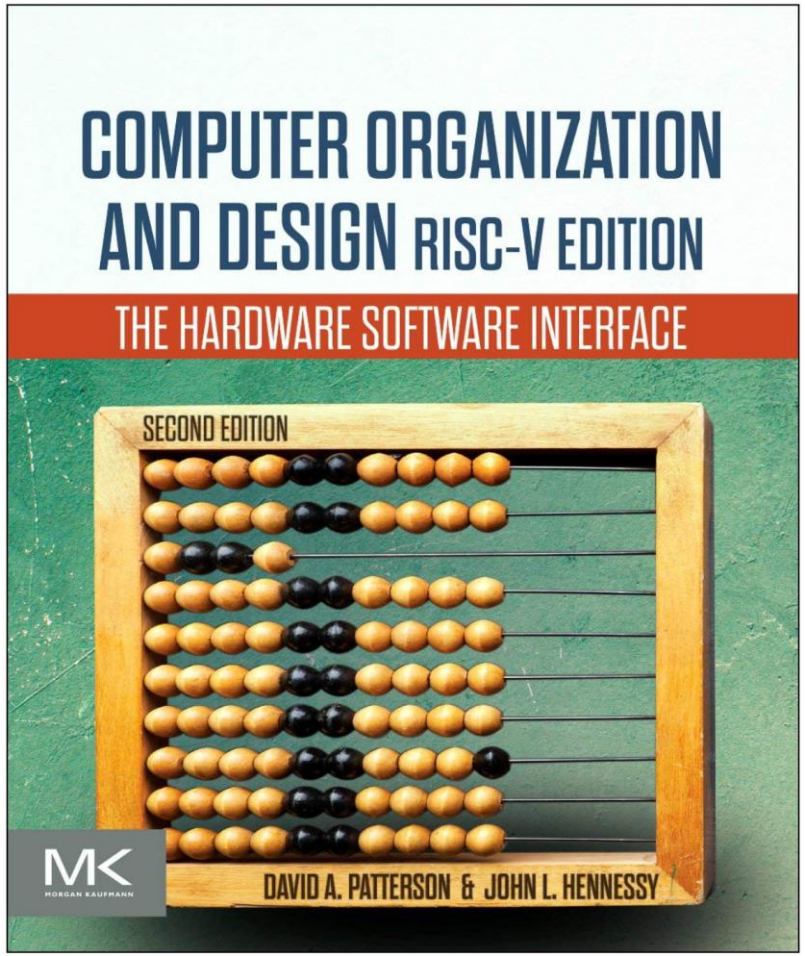
  - https://www.pearson.com/en-us/subject-catalog/p/modern-operating-systems/P200000003295



Reference: RISC-V (1)

- Computer Organization and Design: The Hardware/Software Interface (RISC-V Edition)

  - David A. Patterson and John L. Hennessy (Turing Award Recipients in 2017)
  - Second Edition
  - Morgan Kaufmann, 2020
  - http://booksite.elsevier.com/9780128203316/

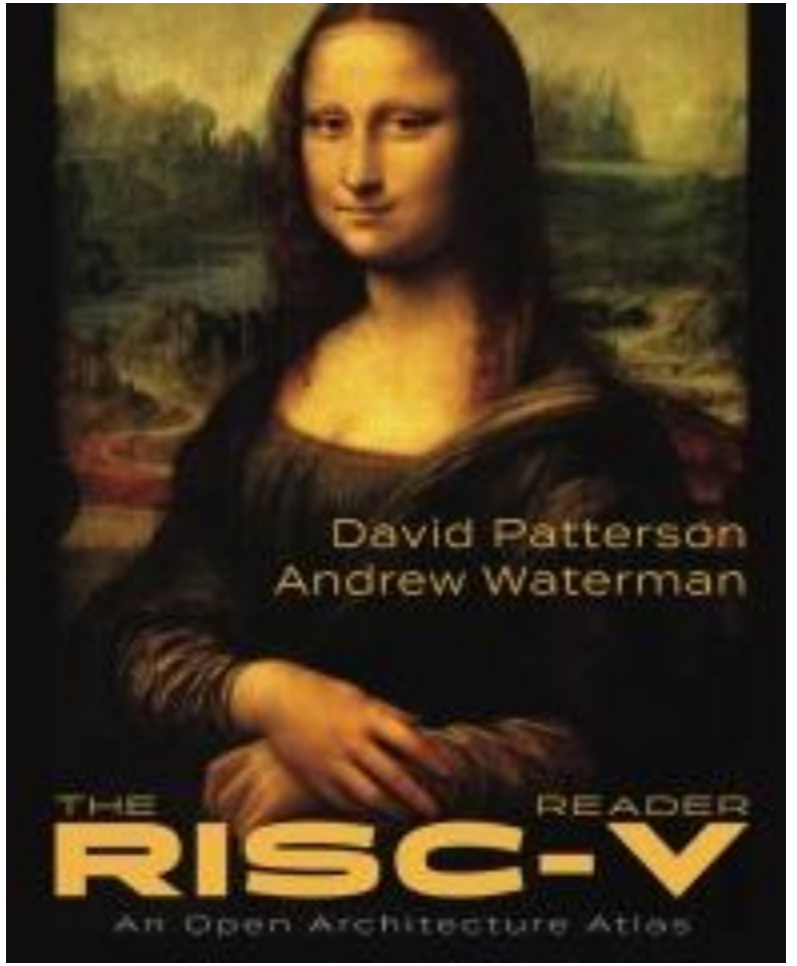


Reference: RISC-V (2)

- The RISC-V Reader:
  An Open Architecture Atlas

  - David A. Patterson and Andrew Waterman
  - Strawberry Canyon, 2017
  - http://riscvbook.com/

  - The free Korean (pdf) version is available



Reference: RISC-V (3)

- https://riscv.org/technical/specifications/
  - Volume I: Unprivileged Architecture (v20260120) ← 권한이 없이도 수행
  - Volume II: Privileged Architecture (v20260120) ← 권한이 있어야 수행





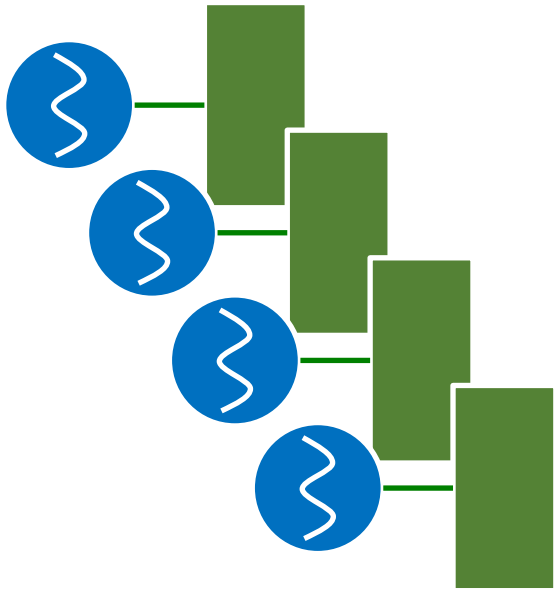
Course Plan

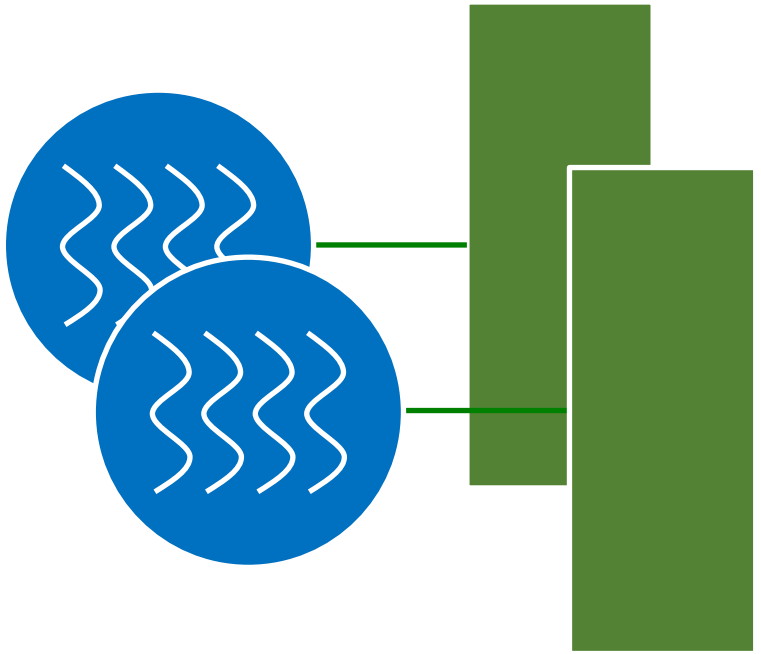
- Lectures
  - General operating system concepts
  - Case study: Linux, xv6

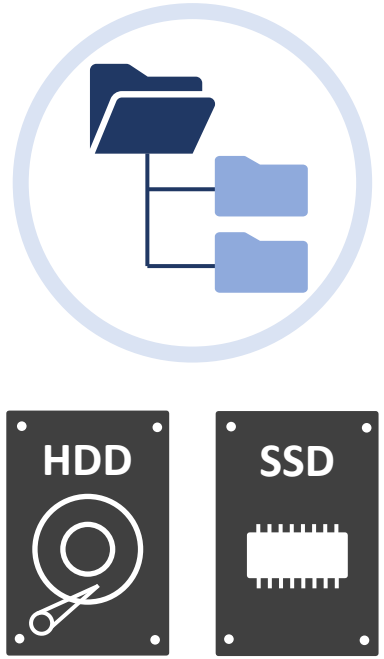
- Hands-on projects
  - Using xv6 instructional OS
  - Based on 64-bit RISC-V architecture

Lectures: Topics

- Virtualization
  - Process
  - CPU scheduling
  - Virtual memory
- Concurrency
  - Threads
  - Synchronization
- Persistence
  - Storage
  - File systems







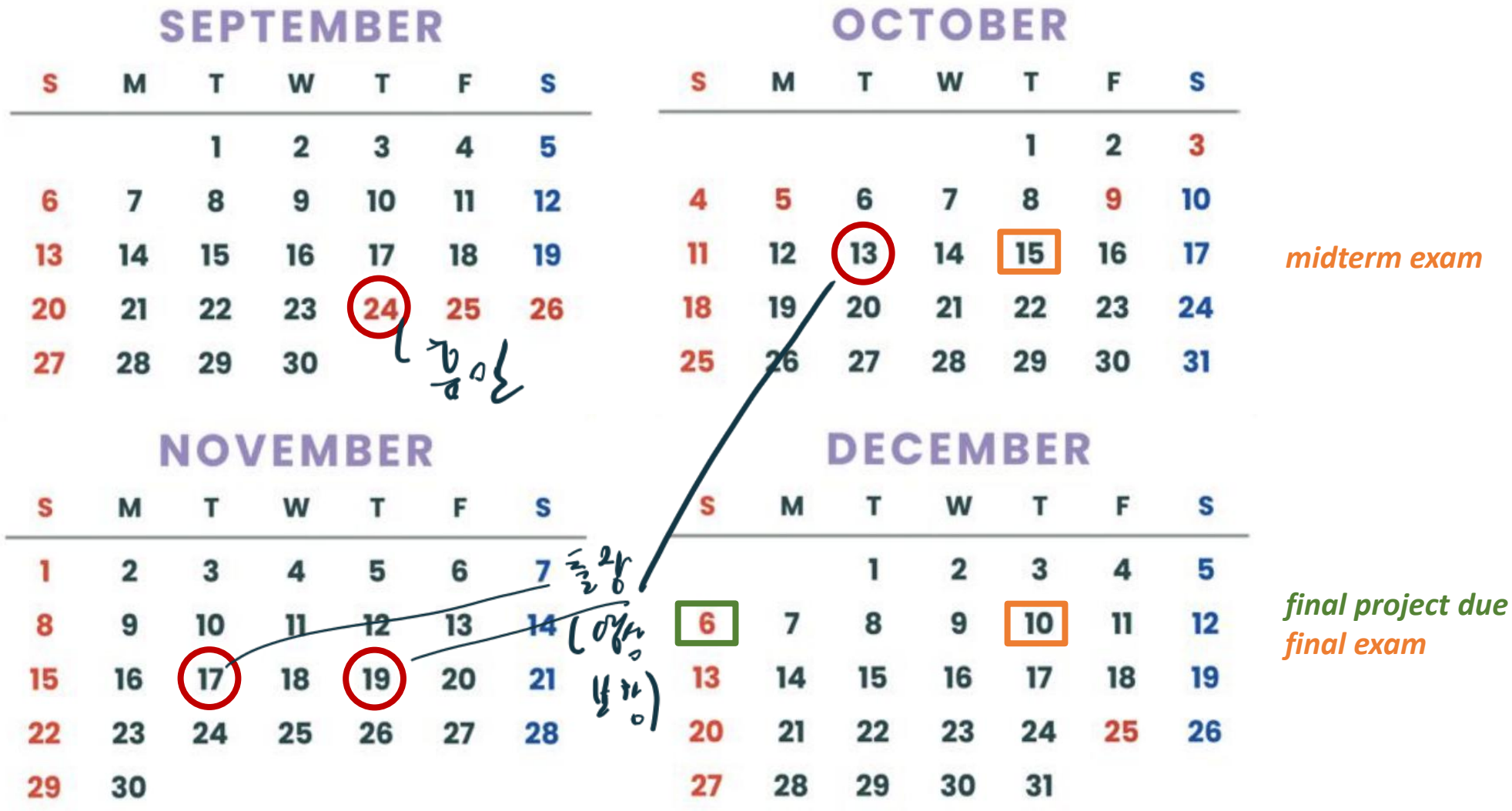
Projects: xv6

- A teaching OS developed by MIT
  - Port of the Sixth Edition Unix (v6) in ANSI C
  - Originally runs on multi-core x86 systems
  - We will use the version that runs on multi-core 64-bit RISC-V systems
- Why xv6?
  - Code inherited from a real, historical OS!
  - Includes working user-level programs and libraries
  - Small: *only* 6K LOCs (vs. 27+ million LOCs for Linux)
  - Easier to install on modern Linux / MacOS systems using QEMU
  - Easier to extend
  - Easier to understand modern OSes such as Linux

Projects Plan

- We are preparing five project assignments
  - The relative weight of each project can vary, typically increasing monotonically
  - Just for your reference: In the previous semester, there were five projects, and their weights were 1%, 2%, 5%, 8%, and 9% for PA1 – PA5, respectively → 10%
- These will be individual projects
- ~~You can use up to 3 *slip* days (Used slip days are irrevocable!)~~
- Lab sessions
  - A separate online class with TAs
  - Project announcement and Q & A
  - Hints & helps
  - …

Schedule

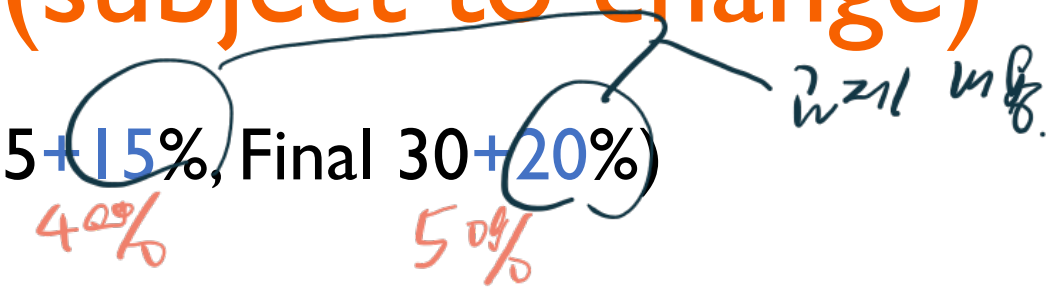


midterm exam

final project due
final exam

Grading Policy (subject to change)

- Exams: 90% (Midterm 25+15%, Final 30+20%)
- Projects: 10%



- University policy requires students to attend at least 2/3 of the scheduled classes. Otherwise, you'll fail this course.
- We are NOT using the electronic attendance system
- If you miss any of the exams, you'll fail this course
- Course withdrawal is only allowed before the midterm exam for students who have taken the prerequisite course "System Programming (M1522.000800)"

Cheating Policy

- What is cheating?
  - Copying another student's solution (or one from the Internet) and submitting it as your own
  - Allowing another student to copy your solution (including publicly posting your solution on Github, etc.)
- What is NOT cheating?
  - Helping others use systems or tools
  - Helping others with high-level design issues
  - Helping others debug their code
- Penalty for cheating
  - Severe penalty on the grade (F) and report to dept. committee
  - Ask helps to your TA or instructor if you experience any difficulty!

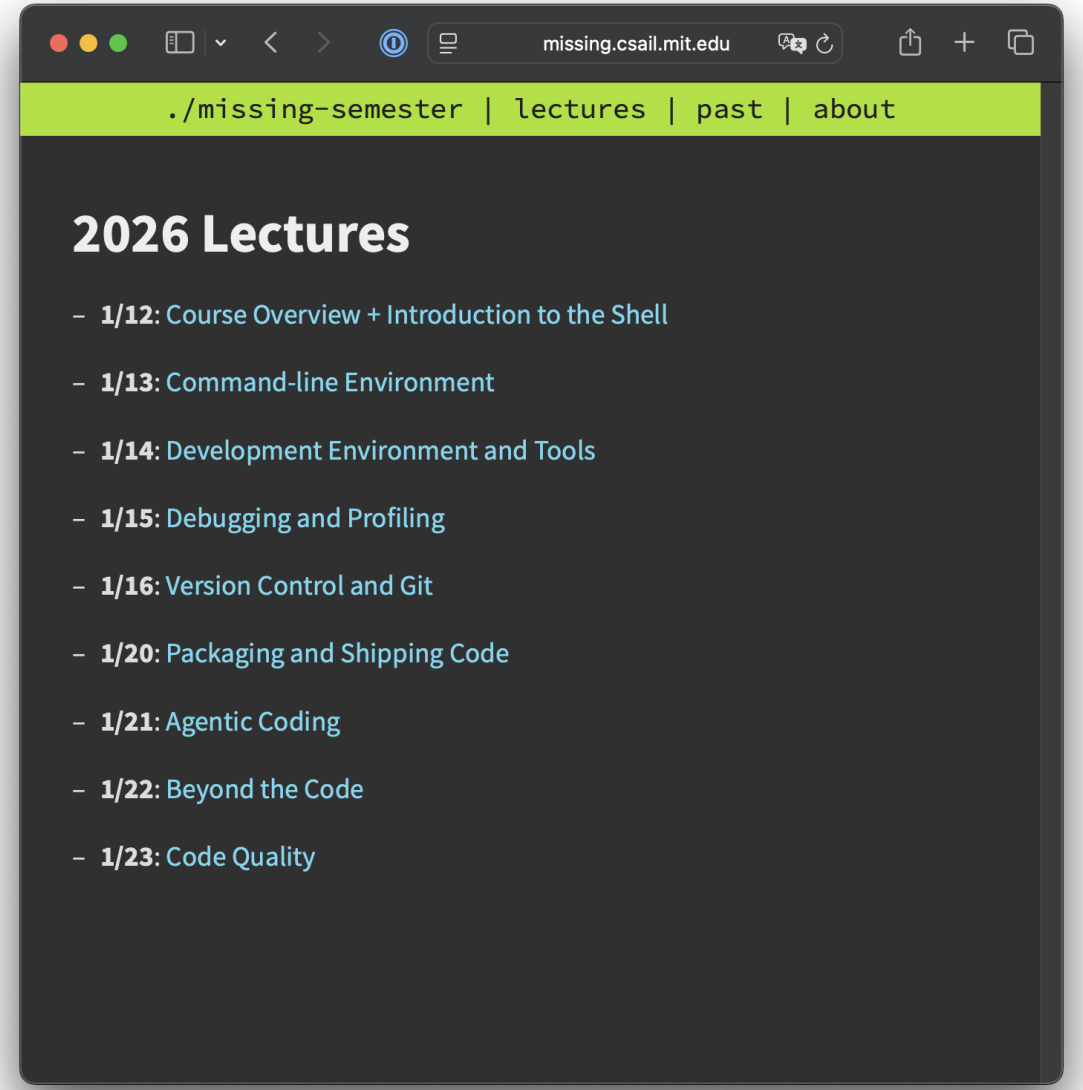
AI Tools Policy for Project Assignments

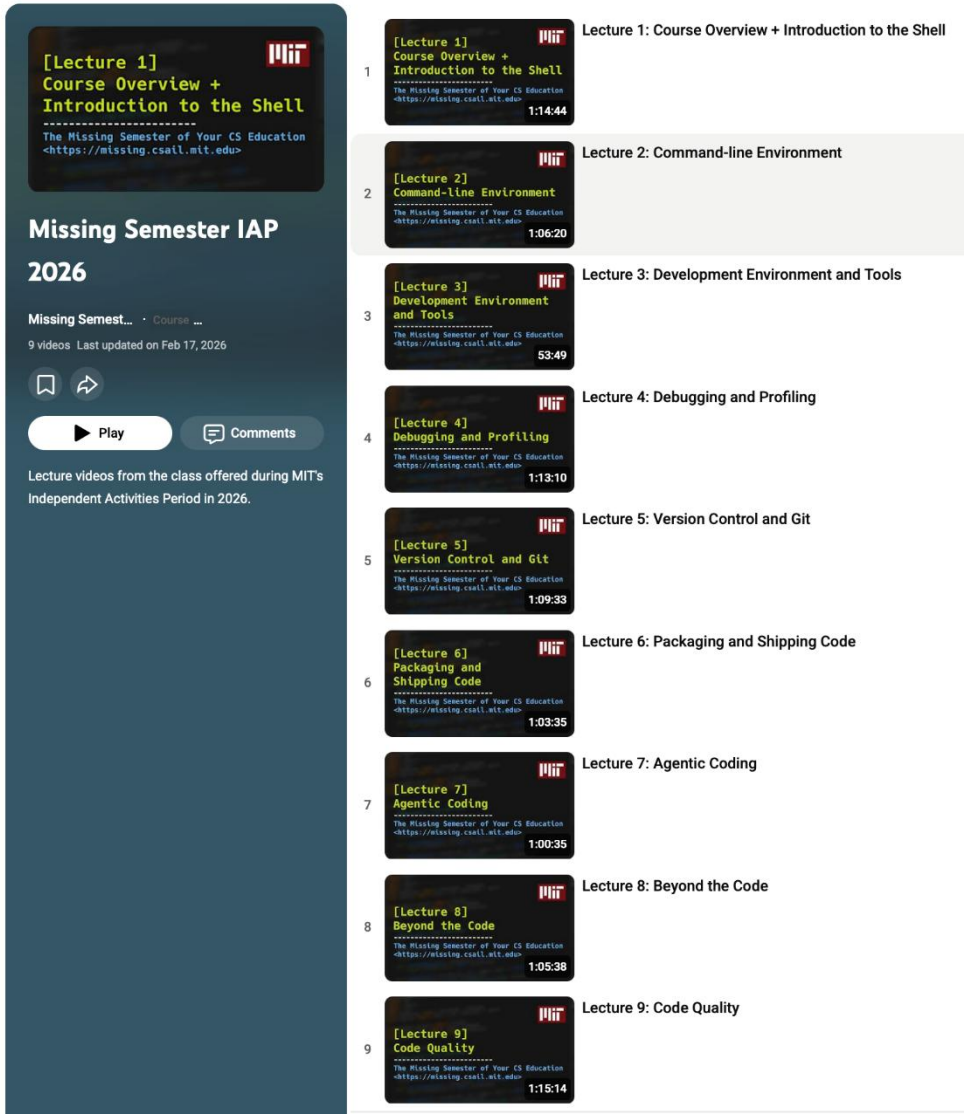
- AI tools are allowed as assistive tools
  - Brainstorming ideas, debugging hints, improving code readability/documentation
- You must understand what you submit
  - You are responsible for every line of code you submit
  - You must be able to explain your design and implementation choices
- Plagiarism policy still applies
  - Submissions with substantially similar code may be treated as plagiarism
- Exams will check your understanding
  - Midterm/final exams will include questions related to the project assignments
  - These questions are intended to assess whether you genuinely understand the concepts and implementation

Summary

- Understanding OS is essential for a broad spectrum of computer systems research & development
  - Embedded systems
  - Cloud computing
  - Distributed systems
  - Security, …
- It has been one of the toughest courses! Use your time wisely
- Please make sure if you're ready to take this course
- Happy hacking!

Missing Semester @ MIT





One More Thing...



1 

[JTJ의 리눅스탐험] 리눅스 설치하기

쓰디 연구소 • 조회수 864회 • 2년 전

2 

[JTJ의 리눅스탐험] 리눅스 기본명령어

쓰디 연구소 • 조회수 343회 • 2년 전

3 

[JTJ의 리눅스탐험] Makefile 활용하기

쓰디 연구소 • 조회수 1.5천회 • 2년 전

4 

[JTJ의 리눅스탐험] Vim Editor 활용하기

쓰디 연구소 • 조회수 497회 • 2년 전

5 

[JTJ의 리눅스탐험] GDB 활용하기

쓰디 연구소 • 조회수 1.8천회 • 2년 전